

A primer on Acceptance Sampling

UGAIN – Black Belt in Six Sigma – Module 3

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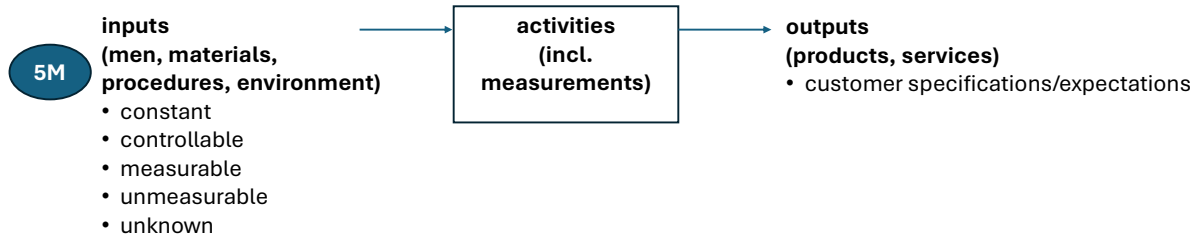
Additional information is available on the notes pages.

Agenda and tentative timing

Subject		Tentative timing (min)
Six Sigma and Quality		25'
Process variation and capability		15'
Common sampling techniques		50'
Acceptance sampling		30'
		120'

6σ is all about business processes and their (continuous) improvement

A business process is an interacting set of activities that produce a product or a service for a (internal) customer.
A formal representation of a process, links process inputs to process outputs.



Business process improvement can be about any aspect of the process: cost, productivity, safety, environmental footprint, quality of the outputs, ... In this contribution we will focus on **quality**.

Any process is subject to **variation**. Process variation leads to variation in the characteristics of the outputs.

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5M: Men, Machines, Materials, Methods, Environment (Milieu).

What is quality?

Common sense meaning

- In the popular sense, identifying quality is purely a judgment call, entirely dependent on the perception of the individuals making the determination. **If a good or service satisfies us, we judge it to be something of quality.** The people who make the satisfaction judgment, may be described as customers. Quality then is a matter of **customer satisfaction**.
- **Webster's International Dictionary** – degree of excellence – degree of conformance to standard – inherent intrinsic excellence of character or type – superiority in kind.

Technical meanings

- **Descriptive:** quality defines some characteristic of a good or service. There is no implication of degree of excellence, just the potential for recording measurable fact.
- **Quality of design:** Design objectives are enunciated by management and are derived from the perceived needs and desires of customers.
- **Quality of specification:** What is the effectiveness of design specifications in realizing the stated objectives.
- **Quality of conformance:** This is the degree of excellence of conformance of the service or manufactured product to the design specifications.

Quality within the context of this lecture

- We stick to the interpretation that quality means the degree of conformance to product/service specifications effectively derived from design objectives that adequately represent customer needs.
- This implies:
 - A product or service has observable/measurable characteristics/attributes
 - For which customer specifications exists (= range of acceptable values, with or without target values)
 - Conformity of a product/service means that the values of the characteristics are within the specifications (better quality means closer to target)
- Examples (for debate)
 - Ceramic tiles – a common manufacturing case
 - Chocolate spread – what about taste?
 - Treatment at the emergency department – quality of service

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- **Ceramic tiles:** product characteristics will relate to dimensions, hardness, color, surface aspect, ... and evidently the absence of defects (cracks, scratches, ...). All features that can be measured and for which unambiguous specifications can be set. The aesthetic features (color, some surface aspects) depend on taste, once agreed upon, these too can be translated into clear specifications.
- **Chocolate spread:** the taste of the spread is (similar to the aesthetics for the tiles) not an absolute feature – the right taste will ensure a proper market share – so it seems important to maintain it over time – however objective measurement as well as setting specifications are a true challenge – how does taste relate to ingredients and process conditions? how do we ensure that taste doesn't change? (Food products have a limited shelf life.)
- **Emergency departments of hospitals** have to deal with random service demands in different aspects: time, nature of injuries/diseases, severity of the problem... Nevertheless also here the quality of service can generally be expressed in terms of performance metrics (measurable/observable attributes of the service). Essentially, people expect an emergency service to help them quickly and effectively. When dealing with quality of services, often reference is made to so-called SLAs (Service Level Agreements). Generally these SLAs list metrics for the performance of the process over a larger period of time and a larger set of outputs. As the word 'agreement' suggests, an SLA is some kind of contract between parties: the customer and the service provider. As such, the strict interpretation of the term SLA may not apply to hospital emergency departments. The performance of an emergency department is rather a matter of public health care. Typical for services is the existence of some aspects, important for the 'customer satisfaction', that may be hard to measure objectively. E.g., how friendly/compassionate was the personnel at the emergency department while treating the patient?

Quality: from a single output to the entire process

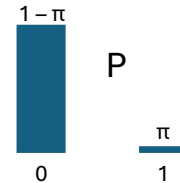
Single output

- Quality as a binary value – an output is conform when **all** quality characteristics are within specification. Expressed as 0 = conform and 1 = defective.
- When target values exist for the quality characteristics, superiority of quality may be derived from comparison to target (the closer to target, the better the quality).

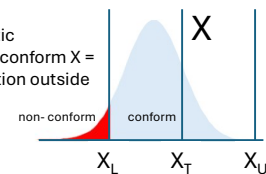
Lot of product - Entire process

- Inherent process variation implies variations in the output characteristics, hence lot/process quality is characterized by distributions.
- The process distribution is being observed as a time series.
- Achieving **stability** of this distribution over time is the ultimate goal of SPC.
- Improving process quality is a major topic in 6 σ and is mainly a matter of **reduction of variability**.

P = (non)-conformity of output
 $E[P] = 0 \cdot (1 - \pi) + 1 \cdot \pi = \pi$
 process with expected fraction non-conform outputs = π
 (all characteristics)



X = output characteristic
 expected fraction non-conform X =
 surface under distribution outside
 specification interval



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The attribute 'conform/defective' covers all characteristics, or at least the set of characteristics that are considered relevant within a given context. For some customers, certain characteristics may not be important. This offers the opportunity for producers to sell product of 'inferior' quality.

Distributions

- P has a binary or Bernoulli-distribution, this distribution shows the probability that the process will produce a conform or a non-conform item. In case of a lot of product, P simply provides the fractions conform and non-conform products in the lot.
- The second distribution shows how a given characteristic varies (either within a large lot or over the entire process). A distribution as shown here is applicable for quantities that can be measured X on a continuous scale (e.g., the length of a ceramic tile). For X we expect the values of this measurement to be between X_L and X_U , with a target of X_T . The actual values delivered by the process (or found in the lot) vary and are sometimes out of specs. The red surface is the fraction of non-conform (related to this characteristic) outputs. (If X would be the only characteristic of interest, then π would be equal to the red surface.)

Being able to measure the variation in the quality characteristics is extremely important. The measurement process is part of the overall process and should be capable to adequately detect the variations in the outputs. That is why MSA (Measurement System Analysis) is an important aspect within the context of 6 σ .

SPC = Statistical Process Control – using control charts to maintain process stability. Also SPC will be dealt with later.

When we look at the distribution of X shown here, we note that the center of the distribution is not at the target value X_T . The process seems not to be well aligned with the specs and in the long run (on average) produces outputs that are a bit too small: $E[X] < X_T$. This phenomenon is called bias or systematic error. Correcting for bias is generally not so hard to do. What is fundamentally more difficult to achieve is the reduction of variability. Here we note that even when the process would be well-aligned with the target X_T , the process distribution is too wide to fit within the specs. So in order to reduce the fraction non-conformities that come out of the process, we need to understand the sources of variation in the process. This is a major topic within 6 σ .

Quality: The system approach of W.E. Deming



Quality

Indispensable for
business success

Quality

Results from the
business system

Key responsibility for
management

- “The basic problem anywhere is quality. What is quality? A product or service possesses quality if it helps someone and enjoys a sustainable market. Trade depends on quality.”
- “The consumer is the most important point on the production-line.”
- “[A] dissatisfied customer does not complain: he just switches”
- “No one can measure the loss of business that may arise from a defective item that goes out to a customer.”
- “The cost to replace a defective item on the production line is fairly easy to estimate, but the cost of a defective item that goes out to a customer defies measure.”
- “Where’s quality made? In the board room, at the top. There’s where quality is made. Jobs are dependent on management’s foresight to design products and services to entice customers, and build a market. To be ready ahead of the customer.⁽¹⁾”
- “The supposition is prevalent the world over that there would be no problems in production or service if only our production workers would do their jobs in the way that they were taught. Pleasant dreams. The workers are handicapped by the system, and the system belongs to the management.”
- “Quality does not just happen: it is built in.⁽²⁾”
- “A bad system will beat a good person every time.”
- “A common disease that afflicts management and government administration the world over is the impression that ‘Our problems are different.’ They are different, to be sure, but the principles that will help to improve quality of product and of service are universal in nature.⁽³⁾”
- “If I could reduce my message to management to just a few words, I’d say it all has to do with reducing variation.”

(1) Next to quality the main driver for business success is innovation.

(2) I.e., in the system.

(3) Understanding and reducing variation.

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One of the key persons in the development of the scientific application of statistical techniques to study and reduce variability in business processes, was William Edwards Deming (1900-1993). As such Deming can be considered as one of the founding fathers of 6σ. In his many writings about the subject he stressed the importance of quality of outputs for lasting business success and the creation and preservation of wealth. For any business to achieve this goal, Deming puts the sole responsibility with management. As management is responsible for the design of the business system and hence the production/service processes. Deming was a gifted speaker and held many management talks over his long career. Many of his quotes, clearly expressing his views on the matter, have become famous.

Besides quality (and the responsibility of management for achieving it), Deming stressed also constant innovation as a key success factor for business:

- “New product and new types of service are generated, not by asking the consumer, but by knowledge, imagination, innovation, risk, trial and error on the part of the producer, backed by enough capital to develop the product or service and to stay in business during the lean months of introduction.”
- “Does the customer invent new product or service? The customer generates nothing. No customer asked for electric lights. There was gas and gas mantles, which gave good light. The first electric lights had carbon filaments. They were fragile and inefficient. No customer asked for photography. No customer asked for the telegraph, nor for a telephone. No customer asked for an automobile. We have horses: what could be better? No customer asked for pneumatic tires. Tires are made of rubber. It is silly to think of riding on air. The first pneumatic tires in the United States were not good. The user had to carry with him rubber cement, plugs, and a pump, and know how to use them.”
- A good question for anybody in business to ask is What business are we in? To do well what we are doing-i.e., to turn out a good product, or good service, whatever it be? Yes, of course, but this is not enough. We must keep asking – What product or service would help our customers more? We must think about the future. What will we be making 5 years from now? 10 years from now?
- “The moral is that it is necessary to innovate, to predict needs of the customer, give him more. He that innovates and is lucky will take the market.”
- “Management is prediction.”

Recommended reading:

- Out of the Crisis – 2nd Edition. Kindle Edition. The MIT Press., 2000.
- The New Economics for Industry, Government, Education – 3rd Edition. Kindle Edition. MIT Press., 2018.

The Quality Ladder⁽¹⁾

Management Approach

Quality By Design

Process Control

Product Inspection

Fire Fighting



Statistics Tools

Design Of Experiments

Control Charts

Sampling Plans

Public Relations

What Deming said about it

Learning is not compulsory; it's voluntary. Improvement is not compulsory; it's voluntary. But to survive, we must learn.

Discovery and removal of a special cause detected by a point out of control is not improvement of the process. This only puts the process back to where it should have been in the first place.

- Inspection to improve quality is too late, ineffective, costly.
- Cease dependence on inspection to achieve quality. Eliminate the need for inspection on a mass basis by building quality into the product in the first place.

One gets a good rating for fighting a fire. The result is visible; can be quantified. If you do it right the first time, you are invisible. You satisfied the requirements. That is your job. Mess it up, and correct it later, you become a hero.

(1) Modern Industrial Statistics – Design and Control of Quality and Reliability, Ron S. Kenett – Shalemyahu Zacks, Duxbury Press, 1998.

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Different styles of quality management can be illustrated by the concept of the quality ladder.

The basic philosophy of statistical thinking is the realisation that variation occurs in all business processes and the recognition that reducing variation is essential to quality improvement. Failure to recognize the impact of randomness leads to unnecessary and harmful decisions.

Level I

- When management's style consists exclusively of fire fighting, there is typically no need for industrial statistics methods. Fire fighting has been characterized as an approach where there is never enough time to do things right the first time, but always sufficient time to rework and fix problems once customers complain.
- (Statistics) Random demand for reports driven by fire fighting. New reports address questions such as: How many components of type X did we replace last month?

Level II

- In the early 1900's researchers at Bell Labs developed sampling techniques that provided an alternative to 100 % inspection to determine if a certain production lot or manufacturing batch is meeting acceptable quality levels. These techniques are focused on the product, as opposed to the process that makes the product.
- (Statistics) Descriptive statistics level. Management asks to see histograms, bar charts and averages. Models are not used, data are analyzed in rather basic ways.

Level III

- In 1924 in a memorandum to management at Bell Labs, Walter Shewhart documented a new approach to statistical process control. When a process is able to produce products that meet customer requirements and a system of process controls is subsequently employed, product inspection is no longer necessary. These are prerequisites for JIT deliveries and increased customer satisfaction. No off-line inspection implies quicker deliveries, less testing and therefore lower costs.
- (Statistics) Process focus. Probability distributions are part of the game. The idea that changes are statistically significant or not, is introduced. Some attention is given to model fitting.

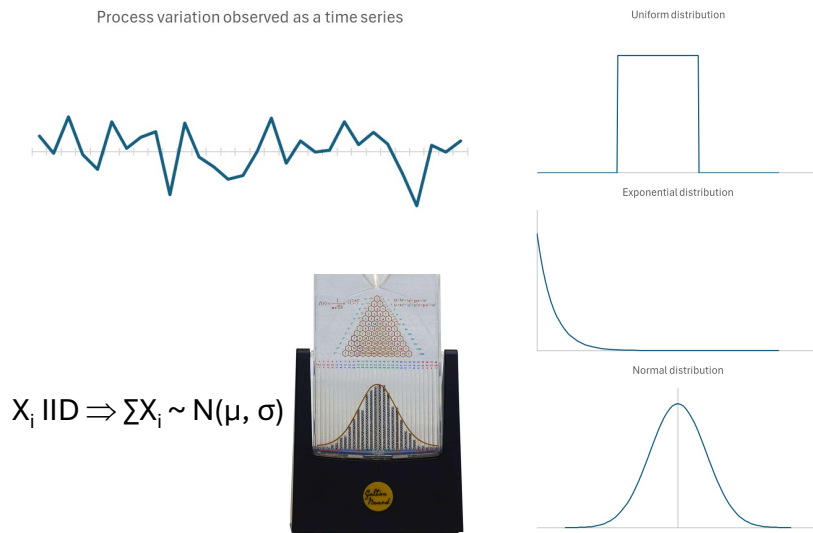
Level IV

- The design of products and processes, both in service and manufacturing, involves quantitative performance measurements. The design of a product and the associated manufacturing process, determines the product's ultimate performance and value. Design decisions influence the sensitivity of a product to variations in raw materials and work conditions. Quality by design is an approach relying on a pro-active management style. Essential elements are quality engineering and quality planning.
- (Statistics) Quality by design. Experimental thinking is introduced. The data scientist suggests experiments, like A/B-testing to help determine which option is better.

Where would you place your organisation on this ladder? The development of maturity in the application of statistics for business processes, is a gradual process that takes both time and consistent management determination. It is considered impossible to skip a rung of the quality ladder.

Describing process variation with distributions

- Process variation is inherently unpredictable, yet a **stable** process will generate outputs that will vary over time between given borders with fixed frequencies
- Of all the different types of distributions to describe process outputs, the so-called **normal distribution** (or Gaussian distribution) is the prevalent one
- The ubiquity of the normal distribution in physical processes is a consequence of the fundamental law, known as the **central limit theorem**
- The basic formulas and calculations within the 6σ framework are based on the normal distribution



“There is no such thing as constancy in real life. There is, however, such a thing as a constant-cause system. The results produced by a constant-cause system vary, and in fact may vary over a wide band or a narrow band. They vary, but they exhibit an important feature, called stability. Why apply the terms constant and stability to a cause system that produces results that vary? Because the same percentage of these varying results continues to fall between any given pair of limits hour after hour, day after day, so long as the constant-cause system continues to operate. It is the distribution of results that is constant or stable. When a manufacturing process behaves like a constant-cause system, producing inspection results that exhibit stability, is said to be in statistical control. The control chart will tell you whether your process is in statistical control.” – W.E. Deming, “Some Principles of the Shewhart Methods of Quality Control”, Mechanical Engineering, 66, pp. 173-177, 1944.

So a stable process is characterized by a fixed distribution of its outputs. When a process is inherently not stable, the distributions will change over time in an uncontrolled way, leading, among other things, to quality issues.

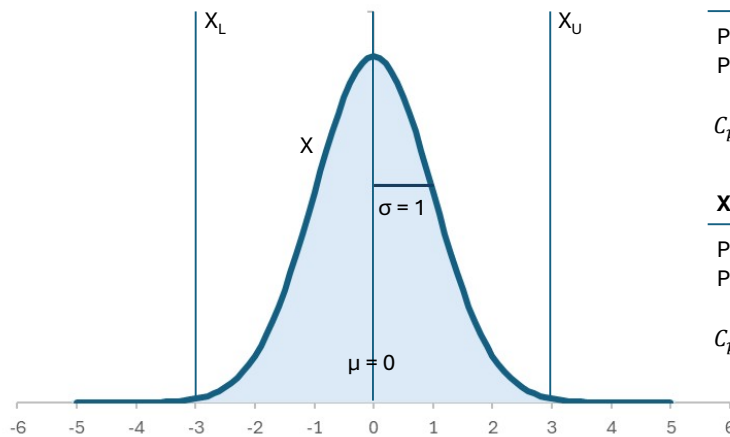
Many physical quantities prove to be normally distributed. The fundamental reason for this is the so-called central limit theorem. If an output is resulting from many different inputs that all have identical (similar) distributions, it will exhibit a normal distribution. This is exactly what happens in a process under tight control. All the inputs are supposed to be at their nominal (target) value, however as variability is unavoidable, the inputs will typically slightly vary around their target. These variations will add up to the typical normal variation of the outputs. A beautiful example to illustrate this fundamental law is the Galton-board.

Question for participants: which distribution on the right of the slide do you think fits with the time series?

In fact the observed outputs of a stable process can be considered as the outcomes of a sampling procedure applied to the **population of all possible outputs the process could ever produce**. Considered in this way, the population is the inventory of all possible process outputs. For most processes, not all outputs are equally numerous, typically the outputs will be centered around the process nominal value, like is the case for a normal distribution.

Note that the process distribution does not capture the time aspect, i.e., how the process outputs are being produced over time. Taking into account the above, for a truly stable process no information gets lost by replacing the time series of outputs into the set of outputs, as there should not be predictive power in the outputs. Such a process is called ‘in statistical control’. When a process is not stable, the time series of outputs will start to contain deviations from the random sequence, e.g., showing auto-correlation between successive outputs. Detecting such deviations is the objective of SPC.

Normal distribution – process capability



$X_U - X_L = 6\sigma$ (see graph)

$P[X_L \leq X \leq X_U] = 99.73\%$ (output conform)

$P[X \leq X_L \text{ or } X_U \leq X] = 0.27\%$ (output defective)

$$C_{pk} = \min\left(\frac{X_U - \mu}{3\sigma}, \frac{\mu - X_L}{3\sigma}\right) = 1$$

$X_U - X_L = 8\sigma$

$P[X_L \leq X \leq X_U] = 99.9937\%$ (output conform)

$P[X \leq X_L \text{ or } X_U \leq X] = 63 \text{ ppm}$ (output defective)

$$C_{pk} = \min\left(\frac{X_U - \mu}{3\sigma}, \frac{\mu - X_L}{3\sigma}\right) = 1.33$$

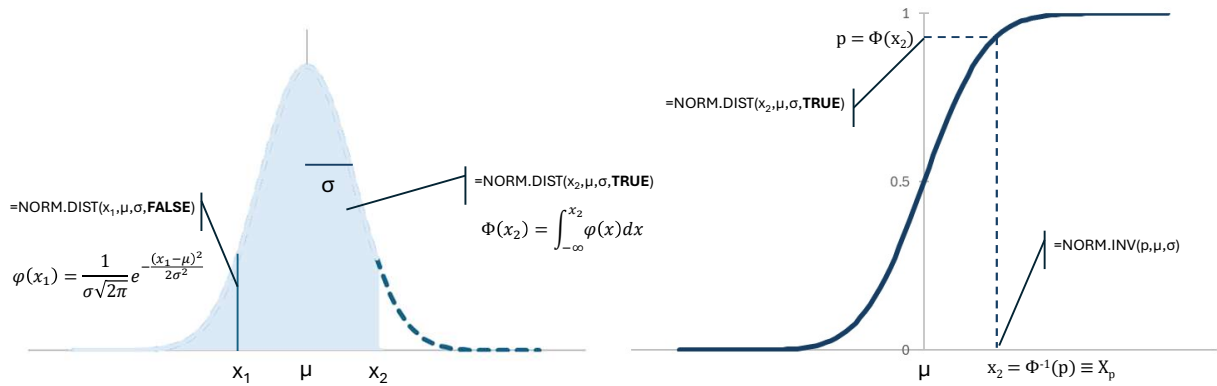
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On this slide is depicted the standard normal distribution, with mean 0 and standard deviation 1. Any normally distributed quantity $X \sim N(\mu, \sigma)$ can be standardized by applying the transformation $Z = (X - \mu)/\sigma$. Hence $Z \sim N(0, 1)$.

When a process is **stable** and generating an output X that is normally distributed and for which we have a specification interval $[X_L, X_U]$, we can predict the fraction of the process output that will be conform (and hence the fraction that will be defective). For a process that generates a fraction f_c of conform products, we say the process is **capable at the level f_c** . Evidently the higher f_c , the better. An equivalent way to quantify the capability of a process is using the quantity C_{pk} . In manufacturing industry (mass production) a common threshold for C_{pk} is 1,33, which means that only about 63 ppm of the process output is defective.

Note: the values μ and σ need to be derived from actual measurements. It should be clear that it is impossible to measure everything about the process. So the actual values for μ and σ used in the calculation of C_{pk} , are estimates of the 'true' process values. The actual measurements we have available form a **sample** of the entire **population** of measurements we theoretically could have taken. To be precise: to calculate the capability C_{pk} , the actual measurements are taken from a limited process time intervals, i.e., μ and σ are based on so-called within subgroup calculations. This will be further explained in the section on SPC.

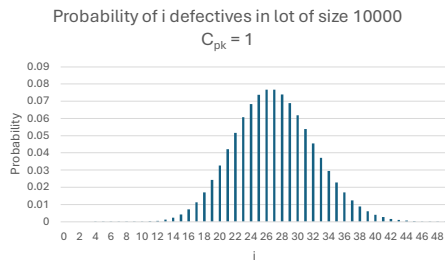
Normal distribution using excel



11

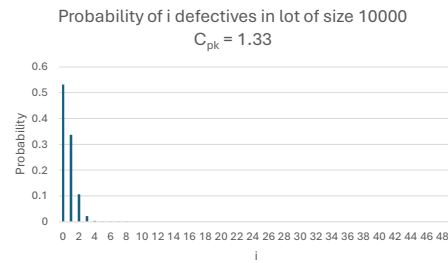
To generate a (pseudo) random sample in excel that is normally distributed, we will use the formula `'=NORM.INV(RAND(),MU,SIGMA)'`. The function `'RAND'` generates a uniformly distributed value between 0 and 1.

Process capability and lot fraction defectives



$$C_{pk} = 1 \Rightarrow \pi = 0.27\%$$

$$N = 10000 \Rightarrow E[i] = 27$$



$$C_{pk} = 1.33 \Rightarrow \pi = 0.0063\%$$

$$N = 10000 \Rightarrow E[i] = 0.6$$

$$P[i] = C_N^i \pi^i (1 - \pi)^{N-i}$$

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What does the process capability (i.e., the process distribution) tell us?

Suppose we have a stable process with a given C_{pk} . Suppose that products are shipped in lots of a given size N . How many non-conformities i can we expect in these lots?

When defectives occur over time without correlation, that is in an independent way, or still stated differently: the fact that a defective item occurs does not impact the non-conformity of any other item produced, the number of defective items in lots will vary in a stable way, described by the so-called binomial distribution of which the formula is given on this slide.

For a process capable at level $C_{pk} = 1$ and lots of size $N=10000$, we see that the number of non-conformities vary between approximately 12 and 44. Does it mean that e.g., 100 non-conformities is impossible? The answer to that question is: in theory 100 non-conformities in a lot can occur, however the probability of this is so low (to be precise $= 2.1 \times 10^{-27}$) that when it would happen, we can rightfully conclude that something must have happened to the process, nullifying its stability, i.e., the process distribution must have changed: the process is no longer capable at the same level as before.

Observe the impact of the process capability on the occurrence of non-conformities in lots. For a process capable at level $C_{pk} = 1.33$, in practice there will never be more than 3 non-conform items in lots of size 10000.

The necessity for lot samples

Given a lot of product coming from some process. How can we derive the quality of the lot, i.e., how can we find out the fraction non-conformities in the lot?

Quality ladder at least level III

- If the process capability is known, we can derive the lot quality from it, i.e., we can deduce the probability of having a lot with given number of non-conformities (as shown on previous slide).
- If the capability is at a sufficient level, we can conclude the lot quality to be acceptable. No inspection is needed.

Quality ladder not higher than II

- Inspection is unavoidable to obtain information about the lot quality
- However 100 % inspection (= screening), may not be feasible because
 - this would take too long
 - this would cost too much
 - testing is destructive
- Taking (a) sample(s) provides an estimate of the lot (=population) quality.

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Whereas it is impossible to measure everything about a process, 100 % inspection of lots is theoretically possible. However the reasons given on this slide prevent this in practice.

Note that 100% inspection may not provide a precise measurement of the lot quality due to inspection fatigue.

When inspecting a lot by sampling, the lot is the population. Within this population the quality level is well determined (fixed). When we consider the entire process to be the population, one lot is just a part of the process output. The quality of lots may (will) vary and the true quality level of the process (over time, all lots) remains theoretically unknown.

On the origin of sampling

- The English word sample does not reveal a common practice of product quality testing, particularly for food stuffs, contrary to German (Stichprobe) or Dutch (steekproef).
- How well does the sample represent the population in the shown examples?



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Let's have a look at the examples shown here.

- A cheese master checks the quality of a cheese by using a cheese drill.
- A grain sampler (Dutch: graan bemonster speer) is used to extract a small amount from a large silo (or bag).
- The core temperature of a ham is being measured with a needle to reach the centre of the ham.
- A sample of wine is taken from a barrel using a wine thief (pipette).

The general word to denote what is being tested, is **population**. So here the populations are: a cheese, a silo/bag of grain, a piece of meat and a barrel of wine.

What we want is that the sample provides us with useful information about the population, i.e., it should guide us to a proper further use of it. When this is the case, we say that the sample adequately **represents** the population. Here:

- Is the cheese ready for consumption and conform to the expected quality for this type?
- Does the grain meet the expectations in terms of type, grain size, humidity, absence of contamination with toxins, etc.?
- Is the meat sufficiently cooked or would it need some further heating? Typically the core temperature is the lowest temperature and hence directional for the doneness.
- How far are we in the riping process? Is the wine ready for bottling? Is the taste and/or smell OK? Etc.

In the examples of the cheese and the wine, we can safely state that the one sample obtained is indeed showing a strong match with the rest of the population. The obvious reason for this being the homogeneity of the population. In these cases the spacial position of the sample in the population is not important. In the example of the meat, the spacial position of the temperature probe is crucial. For food safety reasons it is essential that the coldest point in a piece of meat reaches a certain minimum temperature. So here the sample will only be representative if the needle is indeed pushed to the centre of the meat.

The most difficult case is that of the grain. Typically the grain sampling spear is pushed vertically deep into a silo (or the loading bed of a truck, a train wagon, ...) from the top of the load. We cannot take for granted that the characteristics of the small amount of grain thus retrieved matches the conditions in other parts of the load. (The same holds true for a bag.) So what does a sample tell us in case the population may not be homogeneous in composition? Can we still speak of representativeness of the sample in such case? This proves to be food for controversy. Let us dig a bit deeper into that.

Estimating the fraction defectives in a lot by sampling – case example

The lot

- $N = 10000$ ceramic tiles, packed in boxes of 10 tiles, stacked on 10 palettes of 100 boxes
- Coming from two production batches: A (60%) and B (40%)
- Fraction defective tiles = π (small, **fixed** but unknown number) – $\pi = D/N$ (D = number of defectives in the lot)
- π will be estimated by calculating the fraction defectives $P (=d/n)$ in a sample of size n with d defectives
- **What is the relation between P and π ?**

Common sampling practices (*)

- **Simple random sampling (SRS):** drawing items from the lot merely by chance
- **Stratified random sampling:** upgrading of the SRS-procedure by taking into account sub-populations (strata)
- **Cluster sampling:** procedure based on the hierarchical structure of the population
- **Systematic sampling** – apply some (partially) non-random rule for selecting items
- **Which is the best procedure?**

* Non-exhaustive list.

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The participants are invited to use the excel-file 'Black Belt in Six Sigma - Acceptance Sampling.xlsm' to actually perform the sampling procedures and analyse the outcomes. In this file the lot (population) is described, i.e., for each tile we know what box the tile belongs to, what palette the box is stored upon and whether the tile is defective or not. In practice the (non-)conformity will of course not be available, as it would require a full screening of the lot. This is exactly what we want to avoid by taking a sample.

Evidently the fraction defectives in the lot (π) should be small for the lot to be acceptable for further use.

Simple random sampling is the well known procedure where items are blindly drawn from a bowl, e.g., like in a lottery game or in sports tournaments.

When a population consists of two or more subpopulations, that may be different with respect to the features under consideration, the SRS-procedure can be improved, by composing a sample that has sub-samples from each stratum. In a SRS the entire sample may come from only one subpopulation, which evidently does not provide information about the other strata (see also non-response bias). So here we could create a sample of 100 tiles consisting of 60 tiles of batch A and 40 tiles of batch B. In both batches the tiles are selected at random.

Constructing a random sample from a population of tiles, packed in boxes that are stored on palettes, may be statistically simple, however not so evident in practice. This will generally lead to quite some manipulation work. In order to avoid this, we can exploit the hierarchical structure of the population. Suppose we want to limit the unpacking of tiles to 2 palettes and 5 boxes per palette, then we will select 2 palettes at random and from each of these palettes 5 boxes (again at random). This yields 10 boxes to be fully inspected.

Suppose the boxes are placed on the palettes in four piles of 25 boxes each, then a systematic sampling procedure could be: take the upper box from the left-front pile and inspect the entire box for defects.

The different sampling procedures have their advantages and disadvantages. Depending on the situation, this will lead to a different choice. E.g., a stratified procedure is generally more accurate than a SRS. Both procedures are non-biased (i.e., provide on average the population fraction defectives π), but the random estimate errors of the stratified procedure tend to be smaller than those of the SRS. The practical advantage of the SRS when compared to the stratified procedure is that it is easier to organize.

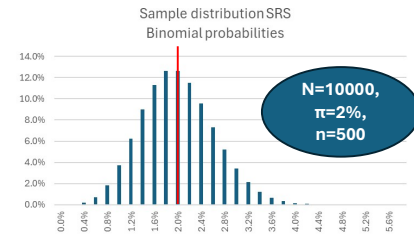
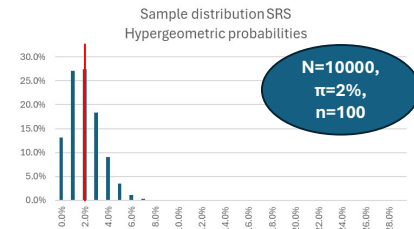
Estimating the fraction defectives in a lot by sampling – case simple random sample

- Let us assume $\pi = 2\%$ (hence $D = 200$).
- How many different samples of size n can be formed out of a lot of size N ? $C_N^n = C_{10000}^{100} = 6,5 \times 10^{241}$
- When each of these samples is equally likely to be selected, the sample is called **fully random**.
- Repeating the sampling procedure will yield different values for P . The sampling procedure introduces variability. This variability will be described by the so-called sampling distribution.
- The correct sampling distribution is the **hypergeometric distribution** – well approximated by the **binomial distribution** (unless small populations)
- P is distributed around π , with $E[P] = \pi$ (P is an **unbiased** estimator of π)
- With increasing sample size, the sampling distribution takes on a normal shape and its variance σ^2 decreases

$$C_N^n = C_{10000}^{100} = 6,5 \times 10^{241}$$

$$P[d = i] = \frac{C_D^i \times C_{N-D}^{n-i}}{C_N^n} \cong C_n^i \pi^i (1 - \pi)^{n-i}$$

$$\sigma^2[P] = \frac{\pi \cdot (1 - \pi)}{n} \text{ (binomial)}$$



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So here we assume the fraction defectives in the lot π to be 2%. Remember that in reality π is unknown and we want to estimate π by taking a sample from the lot.

Even with moderate values for N and n , the number of different samples that can be generated from the lot, is really astronomical. According to cosmological models, the origin of the observable universe (and so the start of time) is situated around 13.8 billion years ago. Hence the number of seconds elapsed since the beginning of the universe is roughly equal to 4.3×10^{17} . So even if constructing one sample would take only 1 second, there would by far not be enough time in the universe to generate all the possible samples.

All these samples have a certain number of defects d . Using the hypergeometrical formula for $P[d]$ on this slide, one can easily calculate how many of the samples exactly have the same fraction (2%) defectives as the population. For $n=100$, this proves to be roughly 1.8×10^{241} . Again a huge number, but quite a lot smaller than the total amount of possible samples. The ratio being 27.5%. So only a minority of the samples give the exact same fraction defectives as the lot. Using a sample will generally produce an error.

As there are 200 defectives in the lot, there will be samples of size 100 that only contain defective items. How many? This proves to be about $9,05 \times 10^{58}$. Also this number is astronomical. However in reality such samples are never observed, as the ratio to the total number of samples is extremely small: around $1,4 \times 10^{-183}$.

Within a lot the fraction defective items π is fixed. It is the sampling procedure that introduces variability. Different samples will generally lead to different estimates for π . As the sampling is done in a purely random way, there is no possibility to explain the outcomes of the procedure. The variation is purely random. Just like for distributions describing process outputs, the sampling distribution provides us with probabilities to observe any given outcome. Very important to note is that on average the procedure will yield the population fraction defectives π . **We say that SRS is non-biased.** This is a very important feature of the procedure.

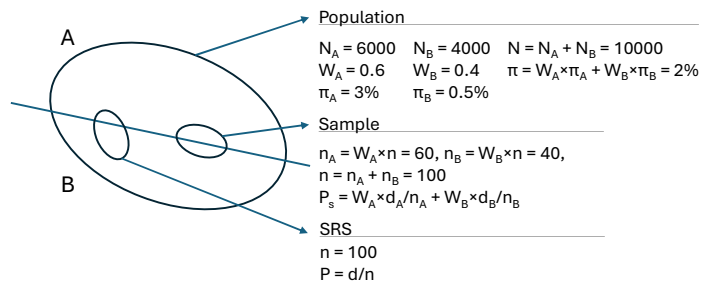
Finally observe what happens to the sampling distribution when the sample size increases. The distribution stays centered around the population value of 2%, but:

- the shape of the distribution starts to look like a normal distribution (this is the result of the central limit theorem, as $P = d/n$ and $d = \sum(0/1)$) (*)
- the sampling distribution becomes more and more concentrated on the population value of 2% (this concentration effect is also clearly demonstrated by the formula for the distribution variance, as n increases, $\sigma^2[P]$ becomes smaller and smaller.

* Evidently the distribution **is not** a normal one. A normal distribution is a continuous distribution, while here the distribution is discrete. However for practical calculation purposes the probabilities deduced from the normal distribution can be used.

Estimating the fraction defectives in a lot by sampling – stratification (1/3)

- The lot consists of 2 batches (A and B) with (very) different fraction defectives: $\pi_A = 3\%$ and $\pi_B = 0.5\%$. There are 6 times more defectives in A than B.
- Suppose the batches can be identified and sampling can be performed out of each batch separately. Instead of taking a RS of size n , we can make two RS, one from each batch and combine these two samples. This is called **pre-stratification**.
- Proportional pre-stratification**: both batches occur in the sample with the same proportions as in the population.
- The sub-samples lead to estimates for π_A and π_B : $P_A = d_A/n_A$ and $P_B = d_B/n_B$, that can be combined into the estimate for π : $P_s = W_A \times P_A + W_B \times P_B$
- Is this any better than a SRS?** Are the sampling errors smaller than those of a SRS?



$$E[P] = E[P_s] = \pi$$

$$\sigma^2[P_s] = \frac{W_A \sigma_A^2 + W_B \sigma_B^2}{n} <$$

$$\sigma^2[P] = \frac{W_A \sigma_A^2 + W_B \sigma_B^2 + W_A W_B (\pi_A - \pi_B)^2}{n}$$

$$\sigma^2 = \pi(1 - \pi)$$

$$\sigma_A^2 = \pi_A(1 - \pi_A)$$

$$\sigma_B^2 = \pi_B(1 - \pi_B)$$

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The formulas on this page can easily be derived from the general formulas:

- $\mu = W_A \mu_A + W_B \mu_B$
- $\sigma^2 = W_A \sigma_A^2 + W_B \sigma_B^2 + W_A (\mu_A - \mu)^2 + W_B (\mu_B - \mu)^2$

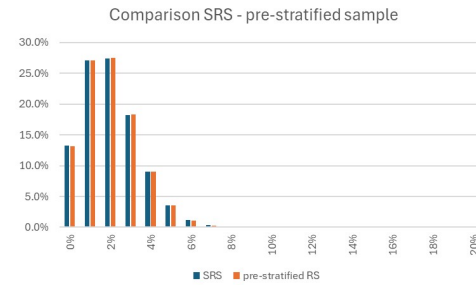
Estimating the fraction defectives in a lot by sampling – stratification (2/3)

- Let's apply these formulas for our case example. Some straightforward calculations lead to the numbers on the right. We note there is barely a difference between $\sigma^2[P_s]$ and $\sigma^2[P]$. Hence the sampling distributions are almost identical.
- Although pre-stratification is theoretically better than a SRS for the estimation of lot fraction defectives, the benefit it offers does not outweigh the disadvantage of being more complex to organize in practice.
- This being said, pre-stratification is worth considering for acceptance sampling by variables (see next page).
- Apart from pre-stratification, there are two other common stratification procedures: **post-stratification** and **optimal stratification**.

$$\sigma^2 = 0.0196 \quad \sigma_A^2 = 0.0291 \quad \sigma_B^2 = 0.004975$$

$$\sigma^2[P_s] = \frac{0.6 \times 0.0291 + 0.4 \times 0.004975}{100} = 0.000195$$

$$\sigma^2[P] = \sigma^2[P_s] + \frac{0.4 \times 0.6 \times (0.03 - 0.005)^2}{100} = 0.000196$$



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Post-stratification is very similar to proportional pre-stratification, however simpler to organize in practice as it does not require the sample to be constructed from sub-samples in each of the strata. Post-stratification starts from a SRS. Normally this sample will contain elements from both strata, however not necessarily proportional to the size of the strata. Nevertheless the elements are divided and averaged per stratum. Then the same formula as on the previous page is being applied, i.e., $P_s = W_A \times P_A + W_B \times P_B$. Post-stratification is unbiased in estimating the lot fraction defectives and will generally lead to smaller random estimation errors than a SRS. Evidently proportional pre-stratification is better (but more complex to organize) than post-stratification.

Stratification reduces the random sampling errors. Proportional pre-stratification is in this respect better than post-stratification. Nevertheless pre-stratification is not always optimal. When the variability (as measured by σ) in one stratum is much larger than in the other stratum, the first stratum should be more than proportionally represented in the sample. The largest reduction in sampling errors for the population fraction defectives π is obtained with pre-stratification according to :

$$n_A = \frac{W_A \sigma_A}{W_A \sigma_A + W_B \sigma_B} \cdot n \text{ (idem for } n_B)$$

$$\text{with } \sigma_A = \sqrt{\pi_A(1 - \pi_A)} \text{ (idem for } \sigma_B).$$

Note that only when $\sigma_A = \sigma_B$ proportional pre-stratification is optimal (*). In our example $\sigma_A \gg \sigma_B$, so stratum A should be represented far more than proportionally in the sample. We can easily calculate that the part of the sample belonging to stratum A should be 78% instead of 60% (and hence 22% of the sample should be from stratum B instead of 40%). Of course in general, the fractions defectives in the strata are unknown and hence also the standard deviations. So optimal stratification is only an option if we have some prior knowledge of the fractions defectives in the strata. This can be done by drawing a prior sample from each stratum, that means increasing the sampling effort.

* In proportional pre-stratification the following formula holds:

$$\frac{n_A}{n_B} = \frac{W_A}{W_B}$$

whereas in optimal stratification this becomes:

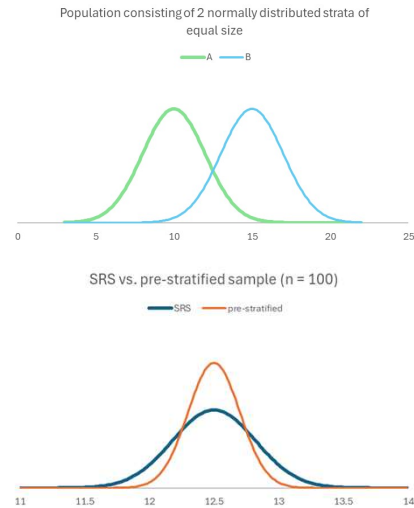
$$\frac{n_A}{n_B} = \frac{W_A \cdot \sigma_A}{W_B \cdot \sigma_B}$$

For those, who like to know how the sampling distribution for the number of defectives d_s (and hence also the fraction defectives P_s) in the stratified sample are calculated, here the formula (again based on binomial probabilities, where to be exact, these should be hypergeometric):

$$P[d_s = i] = \sum_{k=0}^i C_{n_A}^k \pi_A^k (1 - \pi_A)^{n_A - k} \times C_{n_B}^{i-k} \pi_B^{i-k} (1 - \pi_B)^{n_B - i + k}.$$

Estimating the fraction defectives in a lot by sampling – stratification (3/3)

- This page demonstrates the effect of proportional pre-stratification in the case of a population that consists of two equally sized strata (A and B) that are each normally distributed with the same variance, but with different means.
- The means are 10 (A) and 15 (B), the common variance is 4.
- We clearly note that the sample averages ($\bar{X}$, SRS) and ($\bar{X}_S$, stratified) are normally distributed around the population mean (12.5), but with a smaller variance for the stratified sample. So in this case the stratified procedure will produce notably smaller (random) estimate errors, making it worth organizing (despite its greater complexity).



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This page shows an example where we want to estimate the mean μ of a population consisting of two strata (A and B) that are both normally distributed, with the same variance σ_S^2 but with different means (i.e., $\mu_A \neq \mu_B$). Let us further assume that both strata are equal in size (i.e., $W_A = W_B = 1/2$). In this case, based on the formulas of page 19 (slide + notes), we easily derive:

$$\sigma^2 = \sigma_S^2 + \frac{(\mu_A - \mu_B)^2}{4}$$

and

$$E[\bar{X}] = E[\bar{X}_S] = \mu$$

$$\sigma^2[\bar{X}_S] = \frac{\sigma_S^2}{n}$$

$$\sigma^2[\bar{X}] = \sigma^2[\bar{X}_S] + \frac{(\mu_A - \mu_B)^2}{4n}$$

where $\bar{X}$ is the sample average for the SRS and $\bar{X}_S$ the sample average for the proportional pre-stratified sample. The slide shows the results for $\mu_A = 10$, $\mu_B = 15$, $\sigma_S^2 = 4$ and $n = 100$.

The stratified procedure will generate samples that are more representative for the population than simple random samples (of the same size).

Estimating the fraction defectives in a lot by sampling – clustering

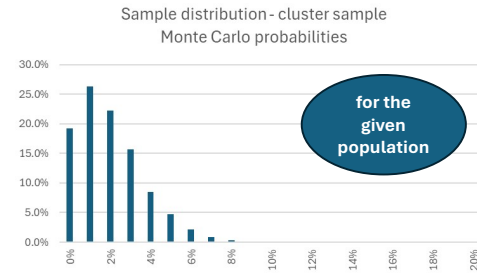
- For a population of tiles, packed in boxes that are stored on palettes, a SRS will generally be laborious to organize in practice.
- To reduce workload, cluster sampling can be considered. The hierarchical structure of the population will be exploited to generate a sample that **combines determinism and randomness**: e.g., pick 2 palettes at random, then pick 5 boxes at random from those palettes and inspect the complete boxes.
- Doing so, the number of different samples drops drastically (compared to SRS), but is still huge.
- The clustering procedure **remains unbiased (*)**, but here tends to generate **larger random errors**.
- Cluster sampling is obviously **sensitive to the distribution of the defectives in the lot**, whereas SRS is not.
- Analytic calculation of the sampling distribution is cumbersome. An outcome is provided by **Monte Carlo simulation**.

$$C_{10}^2 \cdot C_{100}^5 \cdot C_{100}^5 \ll C_{10000}^{100}$$

$$2.6 \times 10^{17} \ll 6.5 \times 10^{241}$$

$$E[P] = E[P_C] = 0.02$$

$$\sigma^2[P] = 0.000196 < \sigma^2[P_C] = 0.000267$$



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* This is however only true if the sample remains random at each hierarchical level. If at one level the sample becomes systematic, non-response bias may occur (see further). For instance: if the 5 boxes from each selected palette are always taken from the top of the palette.

As cluster sampling is partially dependent on a deterministic rule (pick 2 palettes, then pick 5 boxes in each of these palettes), its outcomes will not be the same for lots for which the distribution of the defectives is different, even if those lots have the same fraction defectives. Let's consider an extreme example to make this clear. Let us suppose that all 200 defectives in the lot are concentrated in 20 boxes of 1 palette. For a SRS (of length 100) it doesn't matter where the defectives are situated, so here we have for the probability of drawing a perfect sample : $P[d=0] = 13\%$ (as can be derived from the sampling distribution on page 18). However in the case of a cluster sample, based on 2 palettes and 5 boxes, the probability of having a perfect sample, becomes much larger:

$$P[d_c = 0] = \frac{C_9^2 \cdot C_{100}^5 \cdot C_{100}^5 + 9 \cdot C_{100}^5 \cdot C_{80}^5}{C_{10}^2 \cdot C_{100}^5 \cdot C_{100}^5} = 86\%.$$

In this situation the maximum number of defectives in the cluster sample is 50, whereas a SRS can have 100 defectives (although the probability of this is extremely small).

The fact that a SRS is independent from the distribution of defectives in the lot, is the obvious consequence from the fact that all structural information about the lot is being discarded when drawing a SRS. For a SRS the lot is simply a set of tiles, the notions of box and palette are discarded.

In Monte Carlo simulation we simply repeat the sampling procedures a large number of times and record every time the number of defectives found in the sample. In this way we build up a large population of sampling outcomes, from which we then derive the sample distribution. The result shown here on this page, uses a **particular distribution** of the defectives over the lot, being the one that is shown in the excel-file on the tab sheet 'lot fraction defectives'. Hence the calculated $\sigma^2[P_C]$ shows only the variability introduced by the sampling procedure. As we know that cluster sampling is sensitive to the distribution of the defectives in the lot, we should normally also include this in the Monte Carlo simulation to get a picture of the **total variability**. In other words, the combined effect of the variability of the defectives in the lot combined with the variability of the sampling. However this would make the calculation more complex and the straightforward approach applied in the excel-file, would lead to excessive calculation times.

In Monte Carlo simulation we create in fact some simple digital twin of the sampling procedure. The sampling distribution can then simply be generated by repeatedly observing the behavior of the digital twin.

When the lots would come from a stable production process with a given defective probability π , then in the long run cluster sampling and SRS will provide the same (binomial) sampling distribution. Let us support this statement with an example. Suppose a lot consists of 10 boxes with 10 tiles each. In a SRS of size 20 the probability to finding 3 defective tiles is

$$C_{20}^3 \pi^3 (1 - \pi)^{17} = 1140 \pi^3 (1 - \pi)^{17},$$

whereas in a cluster sample, based on 2 boxes fully inspected, the probability of 3 defectives is

$$2 \times C_{10}^0 \pi^0 (1 - \pi)^{10} \times C_{10}^3 \pi^3 (1 - \pi)^7 + 2 \times C_{10}^1 \pi^1 (1 - \pi)^9 \times C_{10}^2 \pi^2 (1 - \pi)^8 = 2(C_{10}^3 + 10C_{10}^2) \pi^3 (1 - \pi)^{17} = 1140 \pi^3 (1 - \pi)^{17}.$$

Here the boxes do not have a given number of defectives, rather they are described in terms of probabilities containing a given number of defectives. In the long run lots with a favorable distribution of defectives over the boxes will be 'balanced' by lots with an unfavorable distribution. On average this leads to the binomial probabilities of the SRS that is indifferent to the distribution of defectives.

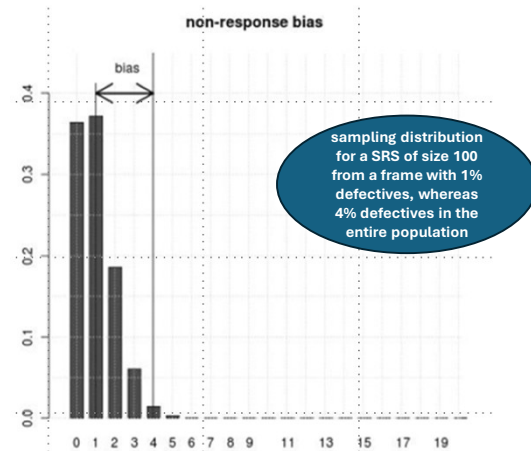
Estimating the fraction defectives in a lot by sampling – systematic sampling

- In a systematic sample, **all randomness is banned** from the procedure. Elements will be selected from the lot (population) based on a **deterministic rule**. For instance (for a sample of size 100 taken from the lot of tiles):
 - Take the first tile from every 10th box (assuming the boxes can be numbered in a meaningful way).
 - Take the first box in the top layer on each palette and inspect the complete box.
- Often the sampling rule is chosen in order to reduce the workload of the procedure. Only elements that are easily accessible are selected. Such approach is known under the name '**convenience sampling**'.
- Like cluster sampling, also systematic sampling is dependent on the distribution of the defectives in the lot (population). In fact the sample distribution is entirely dependent on the variability between lots. Evidently for a given lot, systematic sampling will produce always the same outcomes (and so the sampling distribution 'collapses').
- Typically systematic samples suffer from what is called **non-response bias**. By construction a part of the lot is not accessible for sampling. If this part has a different fraction defectives than the accessible part, this will not be revealed by the sampling procedure.

Estimating the fraction defectives in a lot by sampling – on representativity again

- When sampling errors are acceptably small, the samples may be called representative for the population. These errors are generally the sum of a **random** component (as derived from the sampling distribution) and a **systematic** component (aka **bias**).
- Larger samples lead to smaller random errors, due to the fact that the sampling variance linearly decreases with the sample size.
- However a small SRS may be more representative than a large one that suffers from considerable non-response bias (e.g., a systematic sample, a SRS taken from an accessible part of the population (the so-called **frame**) if this part has characteristic that deviate from those of the entire population.

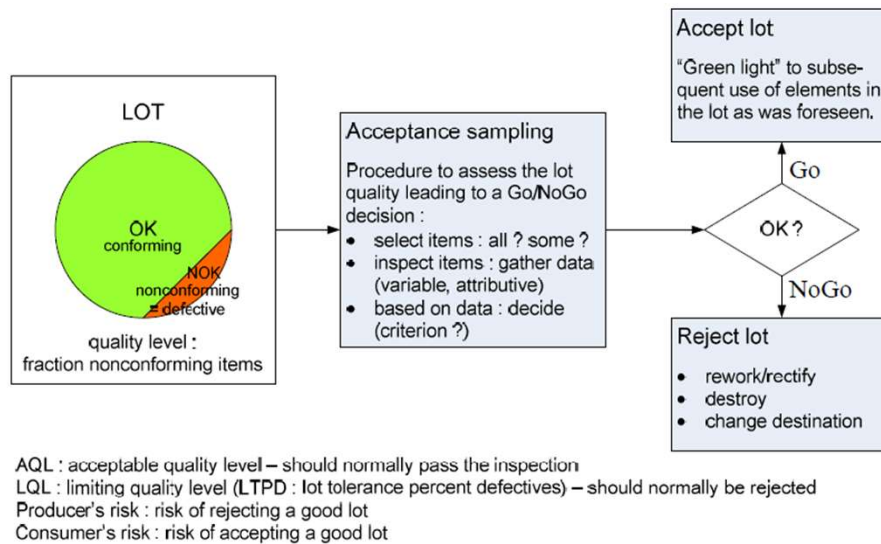
$$\sigma^2[\bar{X}] = \frac{\sigma^2}{n}$$



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If the entire population is not (easily) accessible, a sample will be drawn from the accessible part. This part is called the **frame**. When the characteristics of the frame (such as the fraction defectives in it) are different from the entire population, the sample will not be representative for the population, no matter how big the sample is. The graph shows the sampling distribution for the number of defectives found in a SRS from size 100, taken from a frame containing 1% of defectives, whereas the percentage defectives in the entire population is 4%. In such a situation the SRS will on average under-estimate the fraction defectives by 3%. It is than better to devote the sampling effort on drawing a small SRS from the entire population, than a large SRS from the frame.

The acceptance sampling procedure



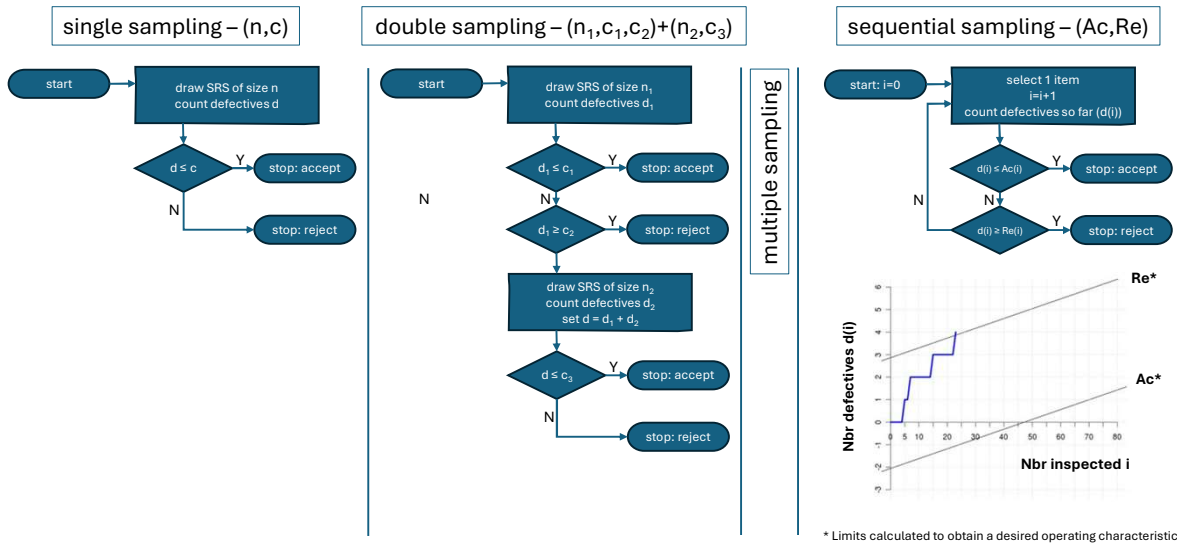
23

On the quality ladder sampling plans are situated on the 2nd level, which is the lowest level of systematic quality assurance methods. Sampling plans are important in those situations where products are received from suppliers (either external or internal) on whom there is no other assessment of the quality level of their production processes. When suppliers have established procedures of statistical process control, then (sampling) inspection of shipments may not be necessary. In such case a supplier will demonstrate that his production process maintains a capability level in accordance with the customer's requirements and as a consequence its products are delivered directly, without inspection, to the production line. But even for certified suppliers it may be a prudent strategy to periodically audit the quality level to ensure that this doesn't drop below acceptable standards (supplier evaluation). A lot is a collection of N elements, i.e., a finite real population of products. The quality level of the lot can be described by the proportion of nonconforming (synonym : defective) items in the lot. A decision needs to be made as to whether a lot has to be subjected to complete inspection (this is often called screening) or whether acceptance can be determined using only a sample from the lot. When using a sample, the sampling plan, the size and the criterion for acceptance need to be determined. The criterion for acceptance of a lot will either be variable or attributive. In the latter case each item after inspection will be classified as either OK or NOK. The test conditions to classify the items have to be well determined, i.e., an operational definition of quality has to be available. Such an operational definition specifies in detail how a quality characteristic will be measured. 100 % inspection will be necessary when items are used in critical or very expensive systems, i.e., when the cost of inspection is negligible compared to the expected cost of failure. In many situations 100 % inspection is not the ideal solution : it may not be possible (destructive testing) or not practical (expensive). In this case inspection will be based on a sample drawn from the lot. **In this situation the decision is taken without knowing the exact quality level in the lot.** Moreover if sampling is based on randomness, repeating the inspection procedure will produce varying results, hence the outcomes of the procedure should be interpreted in terms of **risks**. The procedure should anyhow satisfy the requirement that lots with acceptable quality have a good chance to pass the inspection, while lots with unacceptable quality have little chance to pass the inspection. Acceptable quality is described by the number AQL (acceptable quality level), while unacceptable quality is described by the number LQL (limiting quality level, synonym : LTPD = lot tolerance percent defectives). AQL and LQL should be agreed upon by both the parties involved. The risks involved in the procedure are called : producer's risk (risk of rejecting a lot of good quality) and consumer's risk (risk of accepting a lot of bad quality).

Acceptance sampling plans can be found in the ISO-standards 2859 (attributes) and 3951 (variables). These standards have evolved out of the MIL-STD-105 which in turn resulted out of the pioneering work by Harold Dodge and Harry Romig at Bell Labs in the 1920's.

For now we simply stick to the procedures without going deeper into the statistical considerations. We will come back to these in a further sessions. A document with additional information is available on Ufora for further reading.

Sampling plans for attributes



The sequential procedure shown here is aka the Wald-Wolfowitz procedure. When inspection of an item to decide on its quality takes long, this procedure is not (less) useful. It can be demonstrated that the procedure always ends after a finite number of inspections, i.e., the 'inspection path' will not stay infinitely between the accept and reject lines.

From the graph we can see that the shortest path to reject takes only 3 steps, whereas the shortest path to accept takes about 45 steps.

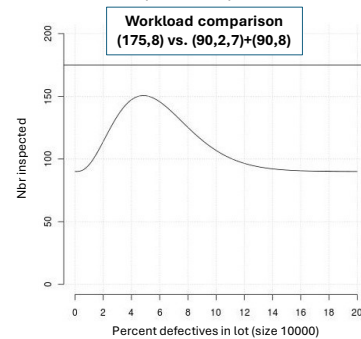
Sampling plans for attributes – risks and efficiency

Risks

- Two sampling plans are said to be equivalent when the **risks associated with the plans are the same**. To be more precise: when both plans have the **same OC-curve** (OC = Operating Characteristic). The OC-curve for a plan shows the probability of acceptance of a lot as a function of the fraction defectives in the lot.
- In particular both for the producer and the consumer, equivalent plans can be interchanged.
- Example: $(175, 8) \Leftrightarrow (90, 2, 7) + (90, 8) \Leftrightarrow$ the sequential scheme (Ac^*, Re^*) shown on the previous page.

Efficiency

- The sampling and inspection **workload** for a single plan (n, c) is proportional to the sample size n . For an equivalent double plan $(n_1, c_1, c_2) + (n_2, c_3)$, the workload is either n_1 (typically smaller than n) or $n_1 + n_2$ (typically larger than n). In the long run, a double plan requires less work than an equivalent single plan, hence is more efficient. It can be shown that a sequential plan is efficiency optimal.



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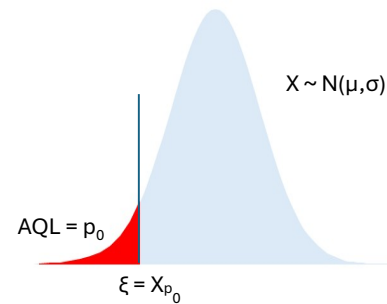
The concept OC-curve is linked to the domain of statistical testing of hypotheses and will be dealt with in a future session of this module.

Single sampling plans for one continuous quality characteristic with lower specification limit (1/2)

- Context:

- a process delivers products for which the quality depends on one continuous characteristic X
- for a product to be conform, the value of X needs to be larger or equal to the lower specification limit ξ
- products are delivered in large lots with values of X (approximately) normally distributed over the lot
- a lot is acceptable for use when the fraction non-defectives is less than or equal to $AQL = p_0$ (e.g., $AQL = 2\%$)

Distribution of the quality characteristic of an acceptable lot (limit case).



An acceptable lot is characterized by a distribution with mean μ sufficiently larger than ξ , taking into account the distribution variation as measured by σ .

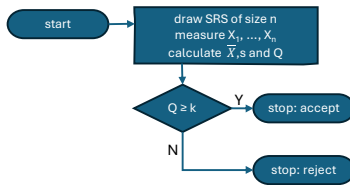
The larger σ is, the further μ needs to be from ξ for the lot to be acceptable.

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For didactical reasons we limit ourselves here to the simple case of a **single plan for one continuous quality characteristic with just a lower (or upper) specification limit**. An example could be the diameter of a shaft, machined to a target value of μ . Due to variations in the process, actual shaft diameters fluctuate around μ in a normal way, with a standard deviation of σ . Shafts with a diameter lower than ξ are no longer acceptable. When the process is stable the process distribution of shaft diameters will be found back in each (large) lot. The part of the distribution lying below ξ is the fraction defectives in the lot. A lot is acceptable for use when this fraction is not higher than AQL.

Single sampling plans for one continuous quality characteristic with lower specification limit (2/2)

single sampling for variables – (n,k)



k is the counterpart of c for single sampling for attributes

$$\bar{X} = \sum_{i=1}^n X_i \quad s = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2}$$

$$Q = \frac{\bar{X} - \xi}{s} \quad Q \geq k \Leftrightarrow \bar{X} - ks \geq \xi$$

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Evidently a good lot (one with fraction defectives at most equal to AQL = p_0 , i.e., $X_{p_0} \geq \xi$) should have a large probability of acceptance, while a bad lot (one with fraction defectives $\gg p_0$) should have a low probability of acceptance.

In order to achieve this, one can set $k = t(1 - \alpha, p_0, n)$ for some small value of α (e.g., 5 %) resulting in a probability of $1 - \alpha$ (95 %) that $\bar{X} - ks \geq X_{p_0} \geq \xi$ leading to acceptance of the lot.

See the notes part of this slide for the formula of $t(1 - \alpha, p_0, n)$.
See the excel file for an illustration of the procedure.

For a normally distributed X , in order to have $P[\bar{X} - ks \leq X_{\beta}] = 1 - \alpha$, k should be set approximately

(for sufficiently large n) to $t(\alpha, \beta, n) = \frac{Z_{1-\beta} + Z_{1-\alpha} \sqrt{\frac{1}{n} + \frac{Z_{1-\beta}^2}{2n} - \frac{Z_{1-\alpha}^2}{2n^2}}}{1 - \frac{Z_{1-\alpha}^2}{2n}}$. This means that with probability $1 - \alpha$

at least a fraction $1 - \beta$ of the population is larger than or equal to $\bar{X} - ks$. Z denotes the standard normal distribution $N(0,1)$.

$$\text{Hence: } t(1 - \alpha, p_0, n) = \frac{Z_{1-p_0} + Z_{\alpha} \sqrt{\frac{1}{n} + \frac{Z_{1-p_0}^2}{2n} - \frac{Z_{\alpha}^2}{2n^2}}}{1 - \frac{Z_{\alpha}^2}{2n}}.$$

Sampling plans for variables – some further remarks

- **Sampling plans for variables are far more efficient than plans for attributes** (so require smaller samples for the same risks). This is a big advantage. The reason for this can be easily understood: translating the actual measurements of the quality characteristic(s) into the attribute OK/NOK (conform/defective), comes with a considerable loss of information.
- The major disadvantage of variable plans is their mathematical complexity, making them harder to explain to uninitiated people.
- Acceptance sampling by variables exists for
 - double and sequential procedures,
 - double specification limits (upper **and** lower),
 - multivariate quality characteristics.

See the professional literature for more information.